

SAL-06 · SALARY AND SKILLS REPORT

Using market data honestly

How to read a pay figure, and how to use one without
misrepresenting it

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PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

— Summary

How to read published pay and skills data without being misled by it: what a median does and does not tell you, why a range without a sample size is not information, why advertised pay is a different quantity from paid pay, how to benchmark a role rather than a title, why a day rate cannot be converted into a salary, and how to bring market data into a pay conversation in a way that survives someone checking it.

— About this document

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Using market data honestly

In one paragraph. This document explains how to read published pay and skills data without being misled by it: what a median does and does not tell you, why a range without a sample size is not information, why advertised pay is a different quantity from paid pay, how to benchmark a role rather than a title, why a day rate cannot be converted into a salary, and how to bring market data into a pay conversation in a way that survives someone checking it.

Who this is for. Practitioners preparing for a pay or promotion conversation; managers and heads of function who have to answer "what should this role pay?"; and anyone about to quote a figure from a salary guide in public.

— 1. What a pay figure actually is

Every published pay figure is the answer to a question that has been asked of some people, at some time, about some component of their pay, and then summarised. Five things are therefore built into it before anyone reads it: **who was asked, how many answered, what exactly was measured, when, and on what basis** — currency, gross or net, and over what period.

Change any one of those and the figure changes, often substantially, without the label changing at all. A figure for "cost engineers" drawn mostly from contractor organisations in one region, measuring base pay only, is a different quantity from a figure with the same label drawn mostly from client organisations across several regions, measuring total cash. Both are honest. Neither is comparable with the other. And neither is comparable with your situation until you have established which one you resemble.

This is not a counsel of despair about market data. It is the difference between using a figure as evidence and using it as a slogan.

— 2. How to read a median

A median is a position, not a typical value. Sort every response from lowest to highest; the median is the one in the middle. If there are eleven responses, it is the sixth. That is the whole definition, and it explains both its strength and its limits.

Why a median rather than a mean. The mean adds everything up and divides. Pay distributions are not symmetric — they have a long upper tail, because a small number of people in any group are paid far above the rest and nobody is paid far below zero. A handful of very high responses pulls a mean upwards until it sits above most of the people it claims to describe. The median does not move when the top response gets larger; only its position matters. When you see a mean quoted for pay, the first question is why the median was not.

What a median does not tell you. Four things, and each of them matters more than the median itself in most real decisions.

- **It says nothing about spread.** Two groups can share a median while one is tightly clustered and the other spans an enormous range. Half the people in a group are below its median by definition, and some of them are a long way below.
- **It says nothing about you.** The median is a property of a distribution, not a claim about any individual in it. "I am below the median" is not evidence of underpayment; it is evidence of being in the lower half, which half of everyone is.
- **It says nothing about direction.** A median is a snapshot. It contains no information about where pay is heading.
- **It says nothing about the shape of the tails.** Whether the upper half is a gentle slope or a cliff changes what a realistic ambition looks like, and the median is silent on it.

Which is why quartiles matter more. The lower quartile is the value a quarter of responses fall below; the upper quartile is the value three-quarters fall below. The distance between them — the interquartile range — is the honest description of where the middle half of a group sits, and it is the number to look for. A source that publishes a median without quartiles has told you where the middle is while withholding how wide it is.

And percentiles are just positions too. The ninetieth percentile is the value ninety per cent of responses fall below. It is not "what good people earn". It is a position in a particular distribution, computed from whoever happened to answer, and in a small sample it is one or two people.

— 3. Why a range without a sample size is not information

A range is two numbers. Without the count behind it, you cannot tell whether it summarises a hundred careers or four.

Worse, "range" is used for two entirely different things and sources rarely say which. A **minimum to maximum** range is defined by its two most extreme responses — a single unusual answer sets one end of it, which is why it widens as the sample grows. An **interquartile range** describes the middle half and is stable. The first is nearly useless for decisions; the second is what you want. They are printed identically.

Five questions to ask of any published figure, in order:

1. **Who was asked, and how were they found?** A volunteer sample recruited through one network is a description of that network.
2. **How many answered this specific figure?** Not the survey's headline total — the count behind the cell you are reading. Response counts fall sharply on pay questions.
3. **What exactly is being measured?** Base pay, total cash, or a rate. If the source does not say, treat the figure as unusable rather than as base pay.
4. **When?** A figure with no reference period is undated evidence.
5. **In what currency, and gross or net?** Converted figures compare nominal amounts. They say nothing about what the money buys, and nothing about tax or mandatory contributions, which differ by country.

A source that cannot answer all five is not necessarily wrong. It is unverifiable, which for the purpose of a decision amounts to the same thing.

— 4. Advertised is not paid

Advertised pay and paid pay are different quantities, and treating them as one is the most common error in this area.

An advertisement is a market offer, written to attract applicants, before any negotiation, before anyone has been assessed, and often before the employer has decided what they will actually pay the person they choose. Whether a range is published at all depends on where the role sits: some jurisdictions require pay information in advertisements and others do not, so the presence, absence and width of an advertised range tells you as much about local law and local convention as about the employer's intent. Never assume the rule that applies where you are applies anywhere else.

Three specific ways advertised data misleads:

- **Selection.** Roles that are hard to fill are advertised repeatedly and prominently; roles filled internally or through networks are never advertised at all. Scraped advertisement data over-represents hard-to-fill and high-turnover roles by construction.
- **The top of the range.** An advertised range's upper end is frequently reserved for a candidate who exceeds the specification, and often goes unpaid. Quoting it as "the market rate" quotes an aspiration.
- **Component ambiguity.** Advertisements mix base, total package, and rates, and rarely say which. Two advertisements for the same job can differ by everything and by nothing.

Advertisement data is genuinely useful for one thing: seeing what employers say they want, in their own words, at scale. That is a requirements signal, not a pay measurement. Used for the first, it is evidence. Presented as the second, it is research theatre.

— 5. Benchmark the role, not the title

Job titles are the worst available classification variable and the most commonly used one. A "senior planner" at one employer integrates cost, schedule and risk across a programme; at the employer next door the same title updates one contractor's schedule. Comparing your pay to a figure attached to your title compares you to whoever else shares the word.

Benchmark instead in three steps.

Decompose the job. Write down what you are accountable for, what artefacts you own, and what decisions are made on your judgement without anyone re-doing them. That is the role.

Level it on behaviour. Autonomy, the scope you own, what you are trusted to decide when the data does not settle it, whose reported numbers you are accountable for, who you influence, and whether you develop method and people. [SAL-03](#) sets out the rubric and its decision rules, including the one people find hardest: you cannot be levelled above the scope you actually own, however senior your influence feels.

Match to the definition, then read the figure. Only after you know which role and level you occupy is a published figure about you rather than about a word.

Two honest complications. First, the same role genuinely pays differently by sector, employer type, contracting model and location, so a match on role and level is a start and not a conclusion. Second, if your job does not fit any published definition — which happens most in small teams,

where one person does four roles — the right response is to say so, and to benchmark the largest component of your time rather than pretending to a clean match.

— 6. Compare like with like — components, and the day-rate trap

Pay is not one number. A comparison is only valid when both sides measure the same components.

Components. Base pay, fixed allowances (site, rotation, housing, transport), variable pay — which divides into what you are targeted at and what you were actually paid — overtime where it exists, employer retirement contribution, and non-cash benefits. Two people with identical base pay can have materially different total positions, and neither of them is being deceptive.

Benefits resist valuation, so do not fake one. Assigning money to healthcare cover, additional leave or a car requires assumptions about local cost and tax treatment that vary by country. Treat benefits as a list you compare item by item, not as a number you add on.

Day rates and salaries are not convertible. This is worth stating carefully, because the conversion is attempted constantly. A contractor's annualised income is not the rate multiplied by a full year:

$$\text{annual income} = \text{daily rate} \times \text{chargeable days worked} - \text{costs borne}$$

Chargeable days are reduced by every day not on assignment: gaps between contracts, unpaid leave, sickness, notice periods and non-chargeable time. Costs borne include insurance, retirement provision the employer does not fund, own training, own equipment, and accountancy or umbrella fees. There is also a risk term that never appears in the arithmetic — the possibility of the assignment ending at short notice — and an employment-status dimension that differs by jurisdiction and which nobody should assume transfers across borders.

There is no general multiplier between a rate and a salary. Anyone who offers you one is offering you an assumption about utilisation, costs and risk appetite, wearing the costume of a calculation. Compare rates with rates and salaries with salaries.

— 7. Using market data in a pay conversation

Market data belongs in a pay conversation as context for your case, never as the case itself. A figure tells your employer what other people are paid; it says nothing about what you have done. The conversations that go well are the ones where the data frames a discussion the evidence about your own work carries.

What to bring.

- The role and level you have matched yourself to, with the reasoning — not the title.
- The specific figure, its source, its reference period, its response count and what component it measures. Being able to say what the number is puts you in a stronger position than the number itself does.
- Evidence about you: what you own, what you decide, what changed because of your work, what you now do that you did not do a year ago.
- An awareness of what the data cannot support, stated by you before it is pointed out to you. This costs nothing and buys a great deal of credibility.

What to avoid.

- **Quoting the upper quartile as "the market".** It is the level a quarter of responses sit above. Using it as the central figure is a misrepresentation, and a manager with the same report in front of them will see it immediately.
- **Quoting a figure whose source you cannot describe.** If you cannot answer the five questions in §3 about it, do not put it on the table.
- **Comparing your base to someone else's total.** The single most common failure in these conversations.
- **Treating your position below a median as an argument in itself.** Half of any group is below its median.
- **Claiming a certification will produce an increase.** No professional body can promise a pay outcome, and the Institute makes no such claim about its own credentials. What a credential evidences is assessed competence against a published standard; what that is worth in a particular role, at a particular employer, is a separate question with its own evidence.
- **Presenting one country's pay-transparency or employment rules as though they applied here.** They may not.

When the data is thin — which, for most specific roles in most specific places, it will be.

Say so. "There is no reliable published figure for this role in this market; here is the nearest defensible comparison and here is why it is not exact" is a stronger position than a confident figure that collapses under one question. It also happens to be true, which is the more durable advantage.

— 8. Reading a skills claim

The same discipline applies to skills findings, where the vocabulary is looser and the temptation greater.

"The most in-demand skill in project controls" is almost always one of three much narrower things: what a self-selected group of respondents said was asked of them; what appeared most often in a set of scraped advertisements; or what the publisher's consultants have noticed lately. Each is a real observation. None of them is demand in any economic sense, which would require vacancies, time-to-fill and wage movement.

Ask which construct is being reported: what people were asked for, what people can do, or the gap between the two for the same person. If the source does not distinguish them, it has not measured any of them carefully. And treat any statement about AI skills as dated from the moment it is published — that part of the picture moves faster than annual reporting can follow, which is a reason to check the reference period rather than a reason to ignore the finding.

— 9. How this goes wrong

The figure outlives its caveats. A number is quoted from a report into a slide, from the slide into a conversation, and by the third hop the sample size, the component and the year are gone. Everything after that is confident and unfounded.

Both sides quote different halves of the same table. The employee brings the upper quartile; the employer brings the lower. Neither is lying and the conversation is now about statistics rather than about the work.

The self-selected comparator. You compare yourself to the two colleagues who talk about pay. They are not a sample; they are the people who talk about pay, which is a specific and unrepresentative trait.

Anchoring on a rate. A contractor's day rate is heard, mentally multiplied by a working year, and used as the standard of comparison. \$6 exists for this, and it will still happen.

Level inflation in the benchmarking. You match yourself against the level you are working towards rather than the one you occupy, and the comparison is off by a level before any figure is read.

Using a hole as evidence. A report suppresses a cell for having too few responses, and someone infers that the role is rare, or undervalued, or in decline. An empty cell means the data was too thin to publish. It means nothing else.

— 10. Checklist — before you quote a pay figure

- [] I can name the source, its publisher and its reference period
 - [] I know how many responses are behind this specific figure
 - [] I know exactly which component it measures — base, total cash, or a rate
 - [] I know the currency and whether it is gross or net
 - [] I have matched my role and level by accountability, not by title
 - [] I am quoting a central measure, not the top of the distribution
 - [] I am comparing the same components on both sides
 - [] I have not converted a day rate into a salary or a salary into a day rate
 - [] I can state what this figure cannot tell anyone, before I am asked
 - [] I am not attaching a pay outcome to a qualification, mine or anyone else's
 - [] I am not assuming any jurisdiction's employment or pay-transparency rules apply elsewhere
 - [] My case rests on evidence about my work; the figure is context, not the argument
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The practitioners who do well in these conversations are not the ones with the best number. They are the ones who can say precisely what their number means, and precisely where it stops — which is the same discipline that makes a forecast credible, applied to a different subject.

— Related

- [SAL-01 – Salary and skills report – framework and methodology](#) — how the Institute's own figures will be produced, and the rules that constrain them
 - [SAL-03 – Role taxonomy and levelling](#) — the role definitions and levelling rubric to benchmark against
 - [SAL-05 – Report template and data tables](#) — the report's structure, including what every empty cell means
 - [CAR-01 – The project controls career roadmap](#) — where a role sits in a career, as distinct from what it pays
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- [ETH-03](#) — [Reporting numbers under pressure](#) — a casebook — the same discipline applied to project reporting

— Sources and standards

No external source is cited, and no pay figure appears in this document. The statistical concepts described — median, quartile, interquartile range, percentile, and the effect of a skewed distribution on a mean — are standard and are explained here in our own words. The role and levelling references are to [SAL-03](#); the Institute's position on certification and pay outcomes is the wording fixed in [00-framework/CANONICAL-FACTS.md](#) §2.1.

— Status and version

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